

Heaven's light is our guide



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Mechatronics Engineering

Assignment

Title: *Advanced Automated Bottling Plant with Integrated Quality Control, Parallel Processing & Fault Recovery.*

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MTE 4102

Automation Sessional

Title: Advanced Automated Bottling Plant with Integrated Quality Control, Parallel Processing & Fault Recovery.

Objectives: This PLC-controlled plant should handle the entire cycle from bottle infeed to packaging, with multiple conveyors, parallel operations, automatic fault handling, and quality inspection.

Automation in Industry 4.0 & Context of Bangladesh

1. What is Automation?

Automation refers to the use of control systems such as PLCs (Programmable Logic Controllers), robots, sensors, and computer-based control algorithms to operate machinery and industrial processes with minimal or no human intervention. The main purpose of automation is to increase efficiency, improve product quality, reduce human error, and ensure consistent operation in industrial environments. In a modern industrial setup, automation allows machines to sense, analyze, and act automatically based on predefined logic or real-time data, ensuring precise and repeatable outcomes.

2. Importance of Automation in Industry 4.0

The concept of Industry 4.0 often called the Fourth Industrial Revolution represents the fusion of digital technology, automation, and data-driven decision-making. Automation is the foundation of Industry 4.0, as it enables intelligent connectivity between machines, humans, and systems through technologies like IoT, AI, cloud computing, and robotics.

The key importance of automation in Industry 4.0 includes:

- **Increased Productivity:** Automated machines operate continuously without fatigue, enabling higher production rates and reduced cycle times.
- **Improved Accuracy & Quality:** Automation ensures precise and repeatable operations, minimizing defects and maintaining uniform product standards.
- **Enhanced Safety:** Dangerous or repetitive tasks can be performed by robots, reducing workplace injuries.
- **Real-time Monitoring & Data Analytics:** Automated systems integrated with sensors and IoT provide real-time data for predictive maintenance and performance optimization.
- **Cost Efficiency:** Although initial costs are high, automation reduces long-term labor and maintenance costs through operational efficiency.
- **Integration & Flexibility:** Automation allows flexible production lines that can adapt quickly to design or volume changes, which is vital in smart manufacturing.

3. Current State of Automation in Bangladesh

The state of automation in Bangladesh is developing but uneven across industries.

- Large sectors like garments, pharmaceuticals, and food processing have begun adopting PLC-controlled systems, robotic packaging units, and automated conveyors.
- However, many medium and small-scale industries still rely on manual or semi-automatic machinery due to cost and skill limitations.
- Pharmaceutical industries (like Beximco and Square Pharma) have implemented partial automation in production, filling, and packaging processes.
- Textile and RMG sectors are slowly introducing automation in cutting, dyeing, and quality control processes.
- In contrast, manufacturing, agriculture, and energy sectors lag behind in adopting advanced control systems or robotics.

4. Challenges to Implementation

- High initial cost of automation equipment.
- Shortage of skilled engineers and PLC programmers.
- Dependence on imported machinery and components.
- Lack of research facilities for local automation development.

5. Possible Solutions to Overcome These Problems

To overcome these challenges and promote automation in Bangladesh, the following solutions can be adopted:

- **Skill Development & Training:**

Establish dedicated automation training centers and university–industry collaborations to train engineers and technicians in PLCs, robotics, and industrial control systems.

- **Government Support & Incentives:**

Provide tax reductions, import duty exemptions, and low-interest loans for industries adopting automation or investing in local R&D.

- **Local Manufacturing of Automation Components:**

Encourage domestic companies to manufacture sensors, PLCs, and robotic components locally to reduce cost and dependency on imports.

- **Public Private Partnership (PPP):**

Foster collaboration between government, academia, and industries to create shared automation labs and research facilities.

- **Awareness & Policy Support:**

Create awareness programs to educate industrial leaders about the long-term benefits of automation in terms of quality, safety, and export competitiveness.

- **Digital Infrastructure Development:**

Strengthen power reliability, internet connectivity, and industrial data networks to support smart factory operations.

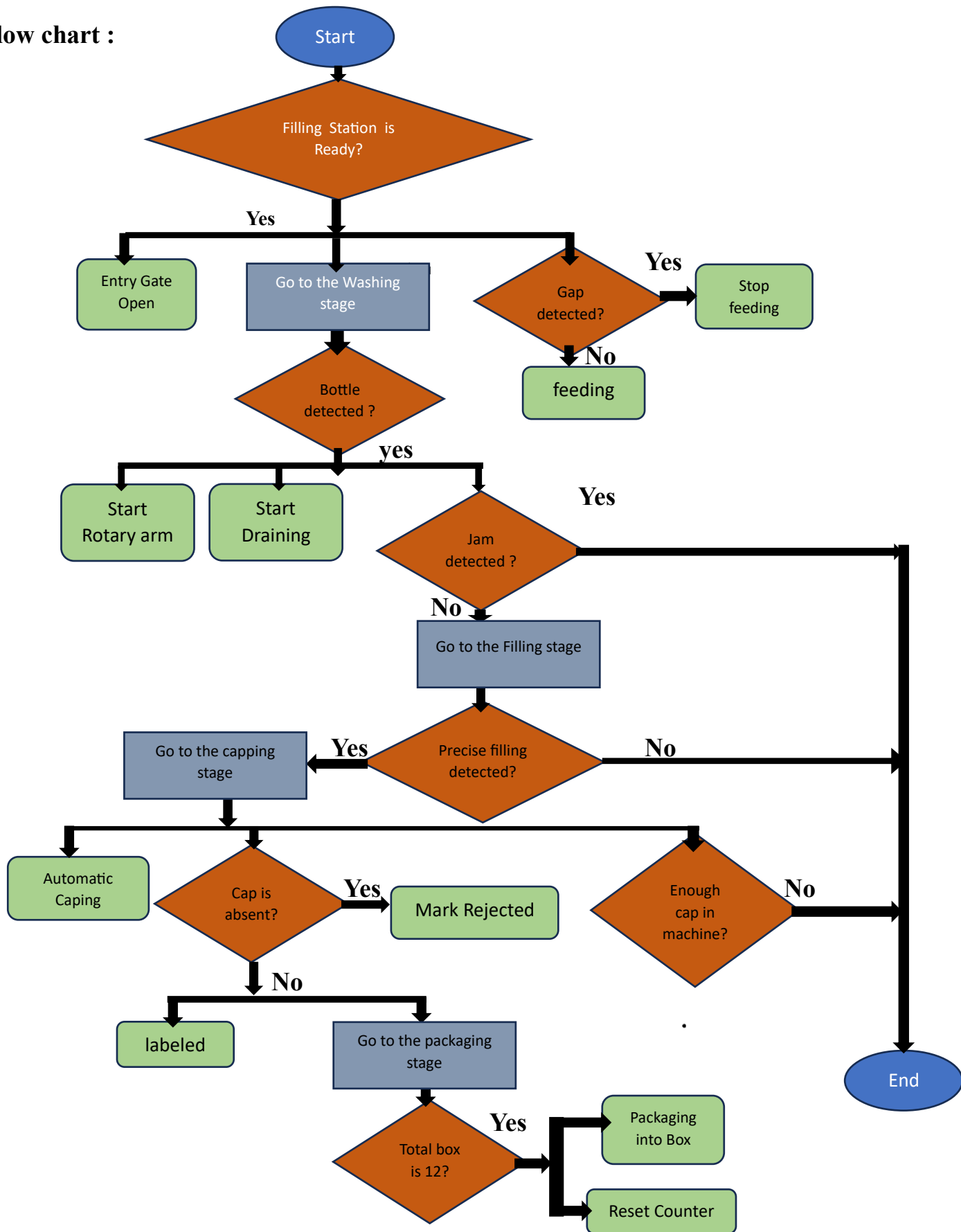
Input Table:

Input Name	Description
Start	Activates or switches on the entire system.
Filling Station Ready	Confirms that the filling station is ready for operation.
Gap Detected	A sensor detects whether there is a gap (presence or absence of a bottle).
Bottle Detected	A sensor detects that a bottle has entered the automatic washer.
Drain Jam	Indicates that the drain is blocked or jammed.
Reached Filling Station	Confirms that bottles have reached the filling station.
Level Sensor	Detects overfilling or ensures precise liquid level in the bottle.
Precise Filling Detected	A sensor confirms that bottles are filled to the correct level.
Cap Absent	Checks whether the bottle has a cap or not.
Not Enough Cap in Machine	Indicates that the machine has insufficient caps for capping.
Label Applied	Confirms that the labeling process has been completed successfully.
Box Counting	Counts how many bottles have been packed into boxes.

Output Table:

Output Name	Description
Entry Gate Open	Opens the entry gate to allow bottles onto the conveyor.
Filling	Feeds bottles onto the conveyor for the washing process.
Skip	Skips the feeding sequence for a position where no bottle is detected (gap detected).
Rotary Spray Arm	Activates the rotary spray arm to wash bottles.
Drain Pump	Drains the used water after washing the bottles.
Jam Detected	Sends a signal to stop the upstream system in case of a jam.
Filling Pump	Starts the pump to fill bottles with the liquid.
No Precise Filling / Overfilling	Sends a signal to stop the system if overfilling or inaccurate filling is detected.
Automatic Capping	Activates the capping mechanism to place caps on bottles.
Mark Rejection	Marks a bottle as rejected due to a detected fault.
Cap Needed	Sends a signal indicating that there are not enough caps in the machine.
Label Applied	Confirms that labeling is completed and the bottle is defect-free for use.
Start Next	Sends a signal to reset the counter to start counting the next batch of 12 bottles.
Palletizes Boxes	Activates the robotic arm to package 12 bottles into a single box.
Load Bottle Arm	

Flow chart :



Sequential Logic:

Start = True

Stage 1: Bottle Feeding & Detection

```
If (Filling Station Ready = True ,  
    Entry gate open =True  
    If(gap is detected = True )  
        Stop feeding = True  
    Else  
        Stop feeding = False
```

Stage 2: Washing Stage

```
If (Bottle is detected = True )  
    Spray rotary arm = True  
    Draining = True  
    If(jam is detected = True)  
        End  
    else  
        go to filling stage
```

Stage 3: Filling Stage

```
If(precise filling detected = True)  
    Go to the capping stage  
else  
    End
```

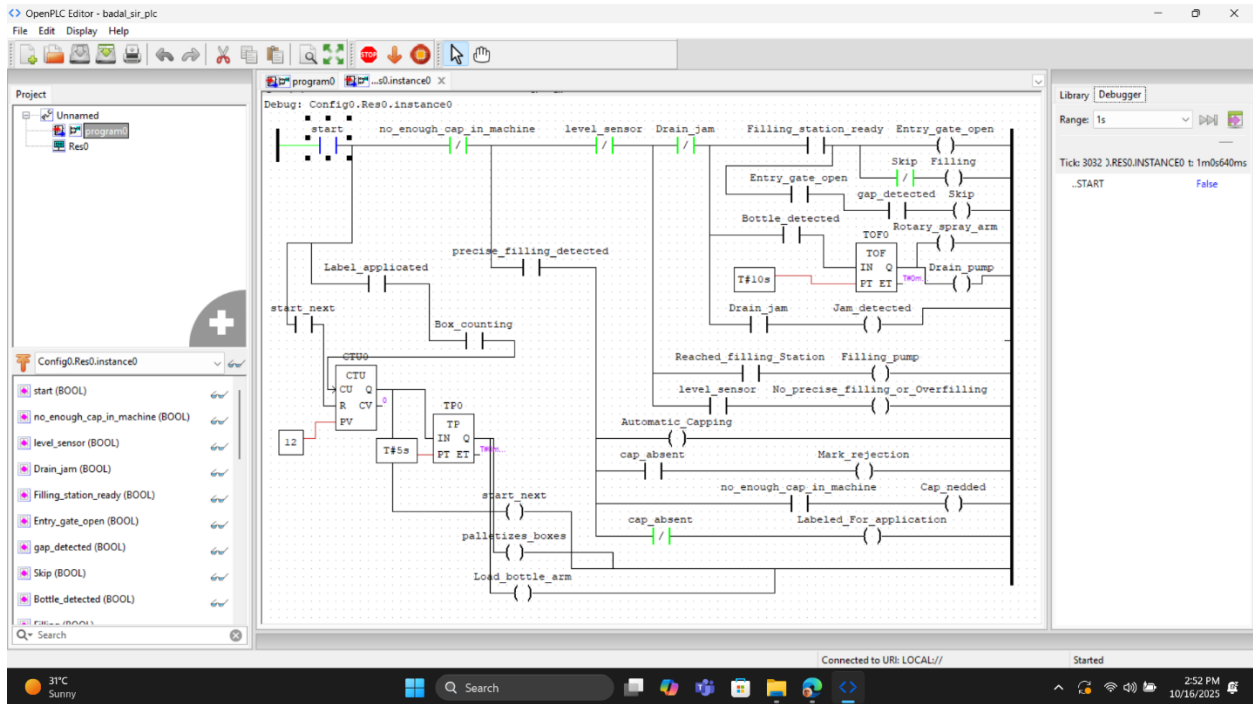
Stage 4: Capping Stage

```
Automatic capping = True  
If(cap is absent= True)  
    Mark rejected = True  
Else  
    Labeled  
If enough cap in machine = false  
    end
```

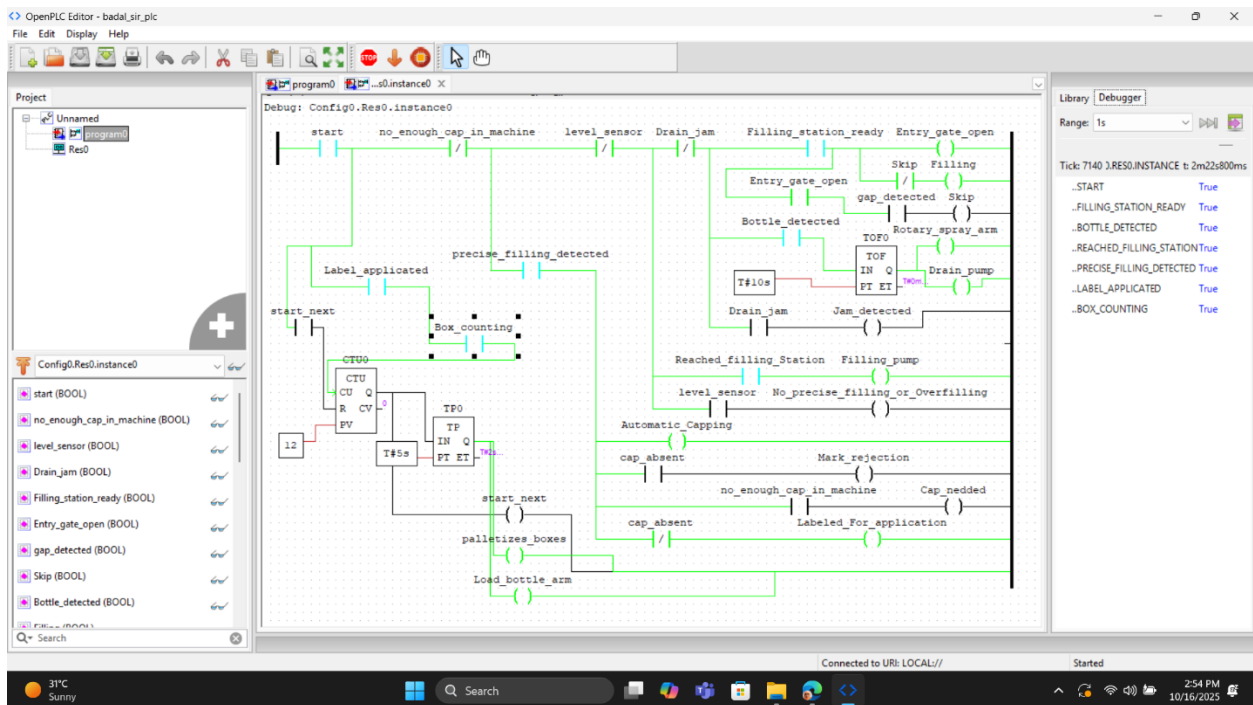
Stage 5: Packaging & Palletizing

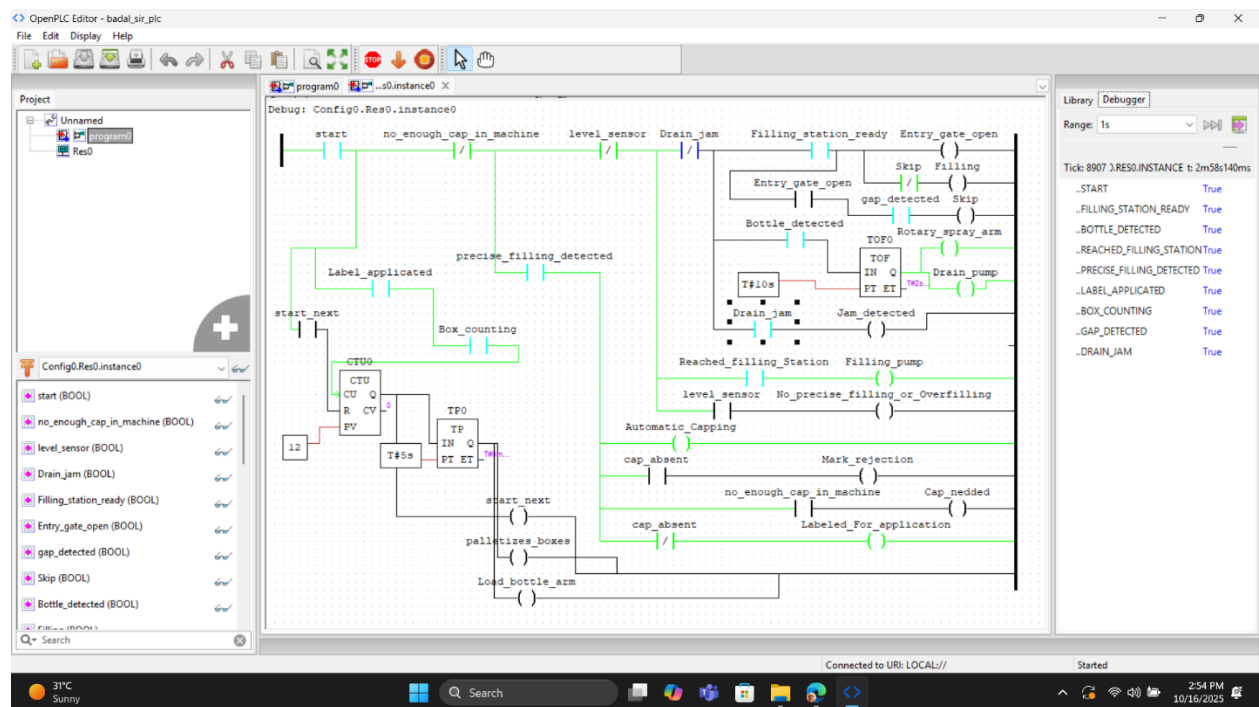
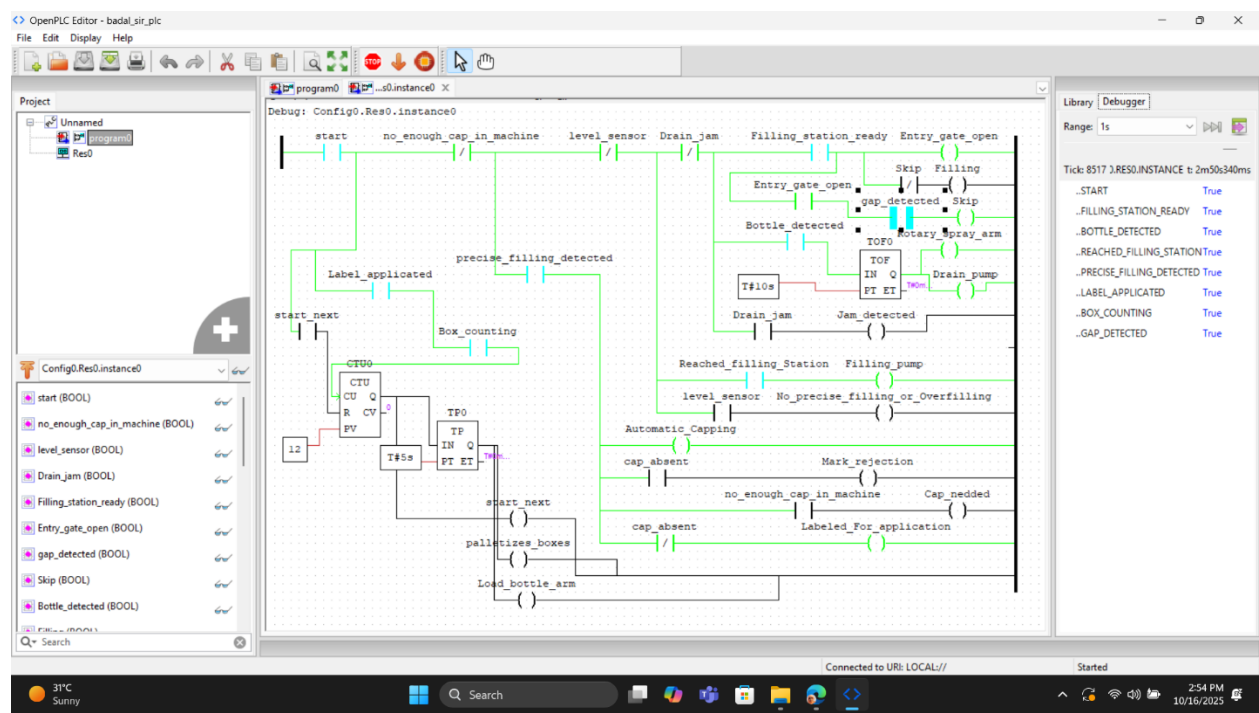
```
If (total bottle counting is 12 = True)  
    Packaging = True  
    Reset Counter = True
```

Ladder Diagram:

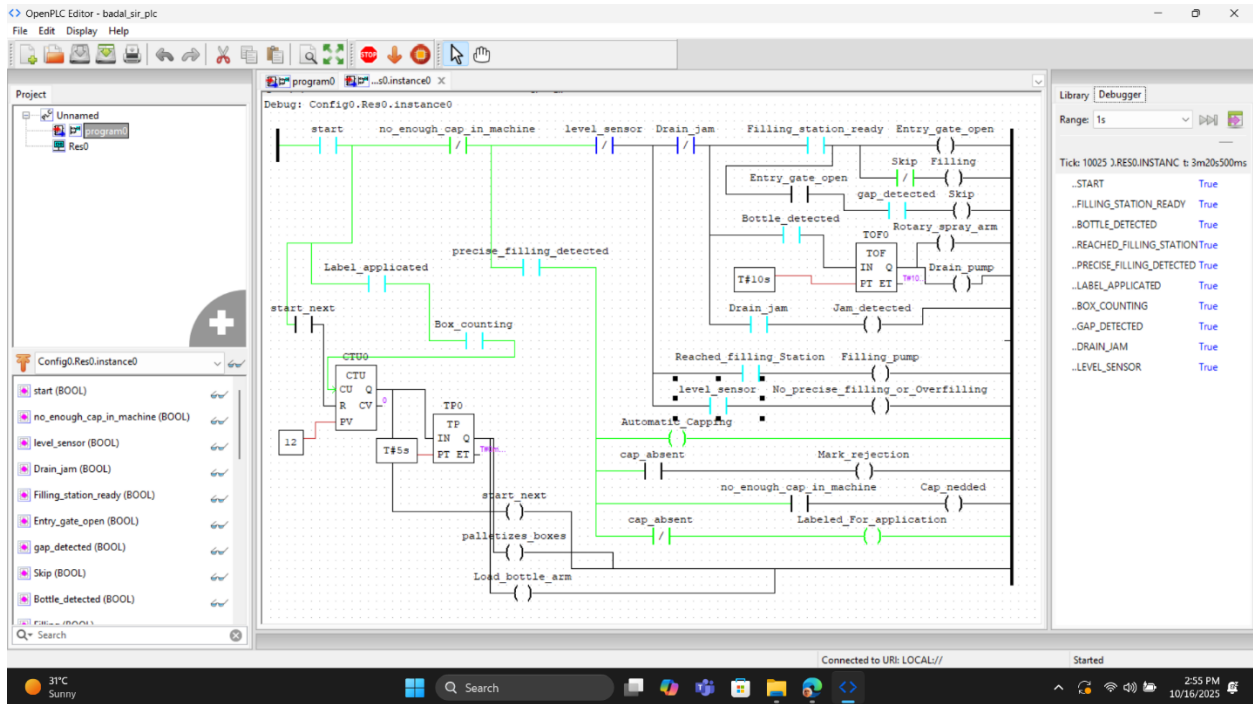


Normal Operation :

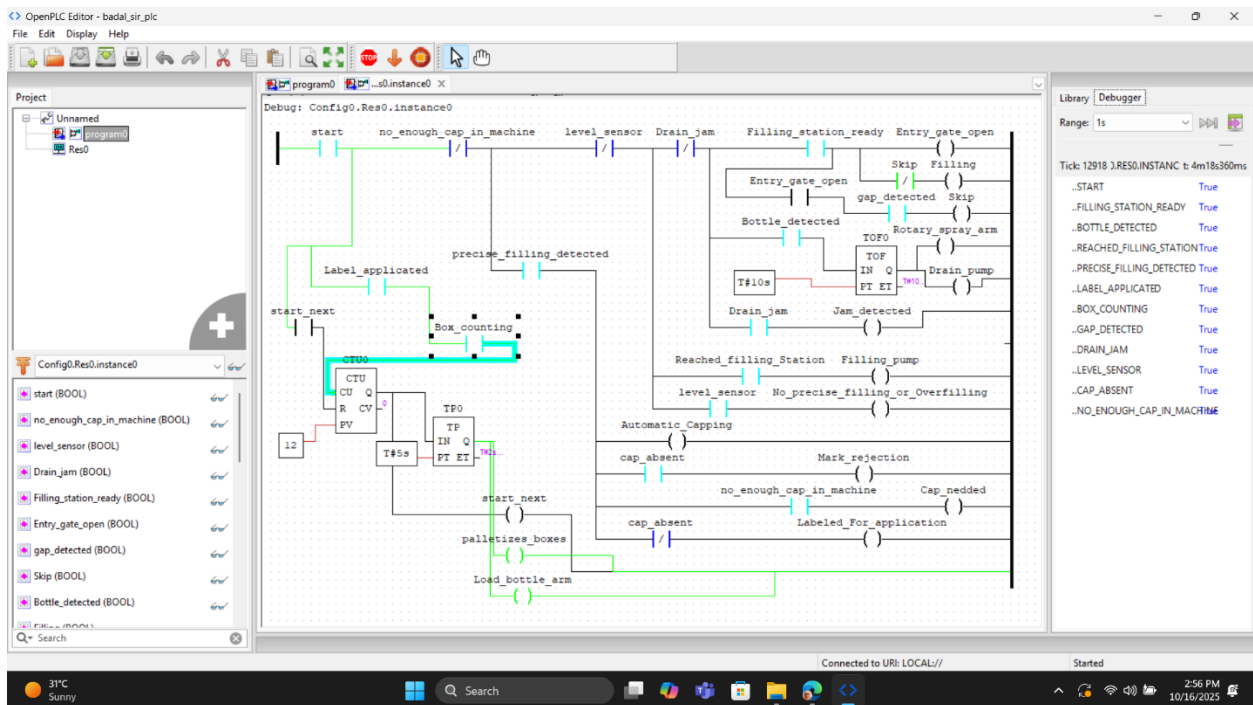




Overflow Scenario:



Cap Feeder Empty Scenario:



The societal and environmental impact of automation:

Automation has greatly revolutionized the modern society through increased productivity, accuracy, and safety in the industries. It reduces mistakes and allows people to focus on more creative and complicated work. However, it also leads to job loss, especially for low-skilled workers, which increases social inequality and economic stress.

From the environmental perspective, automation is advantageous and disadvantageous. On one hand, smart machines and systems can reduce the use of energy, save resources, and utilize resources with an efficient use, with the support of sustainability. On the other hand, production, operation, and disposal of automated machines create electronic waste and carbon.

Therefore, even as automation brought about progress and convenience, society must ensure ethical application, fair transition of jobs, and eco-friendly technology evolution to create a just and sustainable future.

Conclusion:

The successful design and simulation of the Advanced Automated Bottling Plant with Integrated Quality Control and Fault Recovery prove that automation can revolutionize the manufacturing sector. The PLC-based system efficiently manages complex operations through logical sequencing, sensor feedback, and time-controlled mechanisms. The implementation of fault recovery and parallel robotic processing demonstrates the potential of Industry 4.0 technologies to create intelligent, flexible, and self-correcting systems.

This experiment not only highlights the technical aspects of PLC control but also the broader implications of automation in developing nations like Bangladesh. By adopting such technologies, industries can significantly enhance production capacity, reduce human error, and improve product uniformity. However, to ensure sustainability, the country must invest in skill development, local technology innovation, and environmentally friendly automation solutions.

In summary, the project showcases how intelligent automation can bridge the gap between human efficiency and machine precision paving the way for a smarter, safer, and more productive industrial future for Bangladesh.